Diversity and Global Policy: Intergenerational Economic Mobility

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Recap of the Last Lecture

There are large racial and gender earnings and wealth gaps.

Understanding these gaps requires an intergenerational perspective for at least two reasons:

- Parents pass on human capital to their children
- A considerable fraction of total wealth is inherited

Today, we will study intergenerational income mobility based on the following paper:

> *Race and Economic Opportunity in the United States: An Intergenerational Perspective, QJE (Chetty et al., 2020).*

RACE AND ECONOMIC OPPORTUNITY IN THE UNITED STATES: AN INTERGENERATIONAL PERSPECTIVE*

RAJ CHETTY
NATHANIEL HENDREN
MAGGIE R. JONES
SONYA R. PORTER

We study the sources of racial disparities in income using anonymized longitudinal data covering nearly the entire U.S. population from 1989 to 2015. We document three results. First, black Americans and American Indians have much lower rates of upward mobility and higher rates of downward mobility than whites, leading to persistent disparities across generations. Conditional on parent income, the black-white income gap is driven by differences in wages and employment rates between black and white men; there are no such differences between black and white women. Hispanic Americans have rates of intergenerational mobility more similar to whites than blacks, leading the Hispanic-white income gap to shrink across generations. Second, differences in parental marital status, education, and wealth explain little of the black-white income gap conditional on parent income. Third, the black-white gap persists even among boys who grow up in the same neighborhood. Controlling for parental income, black boys have lower incomes in adulthood than white boys in 99% of Census tracts. The few areas with small black-white gaps tend to be low-poverty neighborhoods with low levels of racial bias among whites and high rates of father presence among blacks. Black males who move to such neighborhoods earlier in childhood have significantly better outcomes. However, less than 5% of black children grow up in such areas. Our findings suggest that reducing the black-white income gap will require efforts whose impacts cross neighborhood and class lines and increase upward mobility specifically for black men. *JEL* Code: J0.

Chetty et al. (2020) perform Oaxaca-style decompositions of racial differences in income into components explained and unexplained by parent income.

Overview of the data:

- 2000 and 2010 decennial census
- Tax returns from 1989, 1994, 1995, 1998-2015
- American Community Survey – provides data on hours, wages, education, occupation

They look at child/parent income for cohorts born 1978-1983:

- Parent income averaged over five years
- Child income averaged over two years, at ages between 31 and 37

Before digging into the data, the authors motivate their approach with a simple conceptual framework...

A Simple Theoretical Motivation

Assume each family i comprises one individual in each generation t, such that:

$$y_{i,t} = \alpha_r + \beta_r y_{i,t-1} + \varepsilon_{i,t},$$

where $y_{i,t}$ = the income percentile rank of individual i relative to all other individuals in the same generation t and $r(i)$ = the race of family i.

The mean rank of individuals of race r in generation t is then:

$$\bar{y}_{r,t} = \alpha_r + \beta_r \bar{y}_{r,t-1}.$$

The mean rank of generation $t + s$ is then:

$$\bar{y}_{r,t+s} = \alpha_r \frac{1 - \beta_r^s}{1 - \beta_r} + \beta_r^s \bar{y}_{r,t}.$$

If $s \to \infty$, $\beta_r^s \to 0$ if $\beta_r < 1$, so the long run steady state is:

$$\bar{y}_{r,t} = \bar{y}_{r,t-1} = \bar{y}_r^{SS} = \frac{\alpha_r}{1 - \beta_r}.$$

Quiz

What are the implications of this equation for the evolution of racial disparities?

$$\bar{y}_{r,t} = \bar{y}_{r,t-1} = \bar{y}_r^{SS} = \frac{\alpha_r}{1-\beta_r}.$$

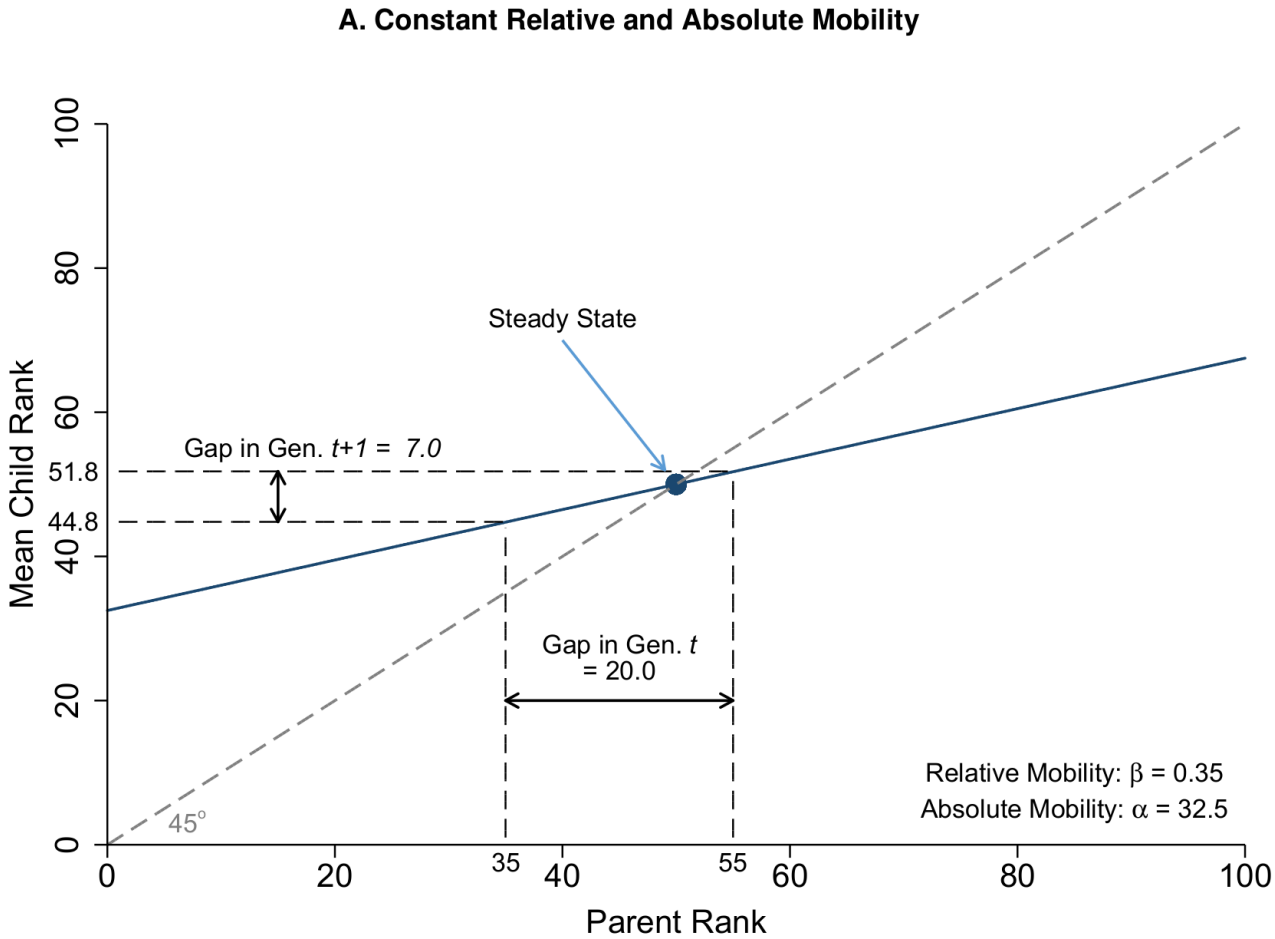
... If α and β are constant?

... If α_r is race-specific but β is constant?

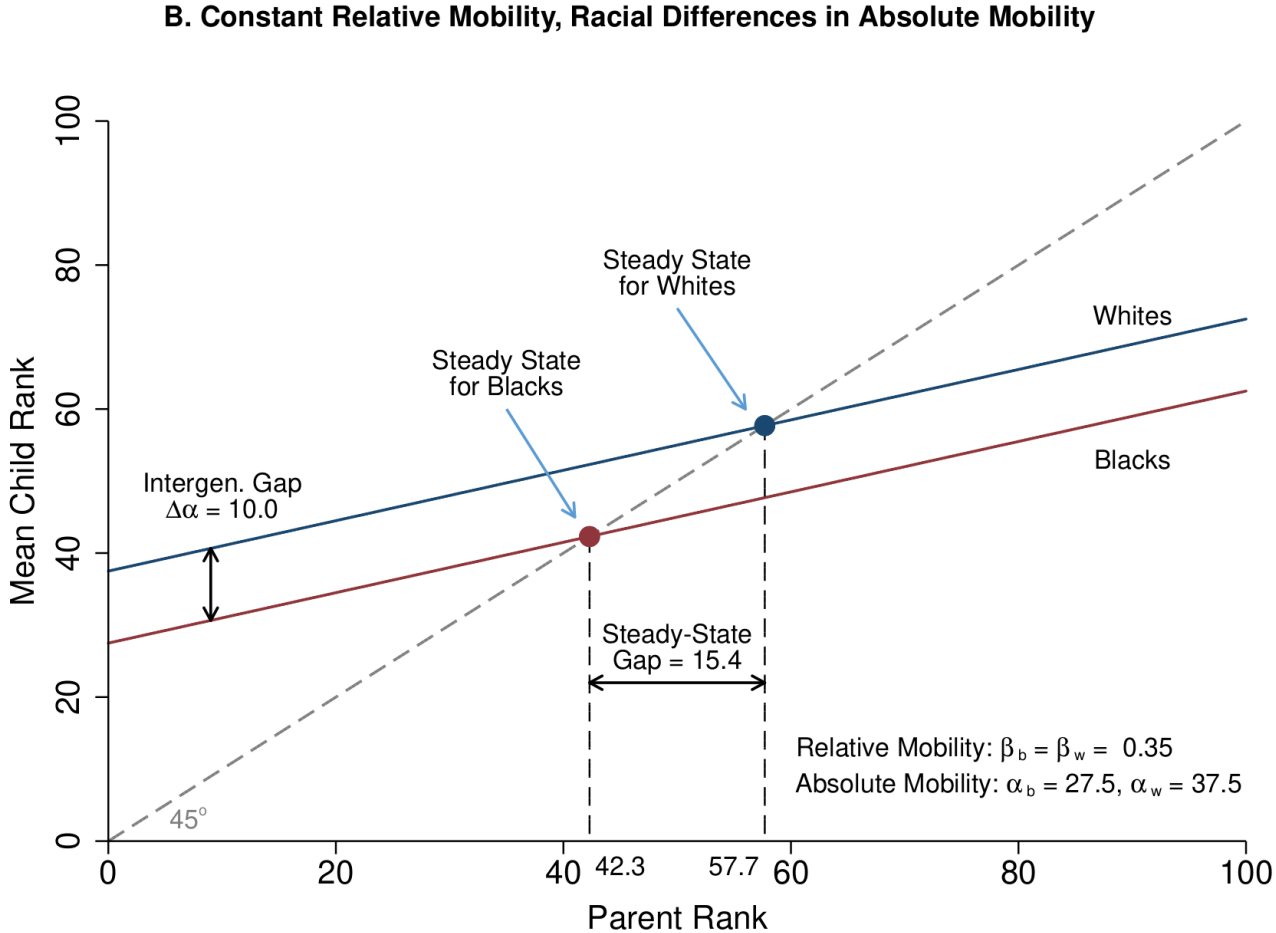
... If both terms are race-specific?

... Do you think the linearity assumption is plausible? Can we test it?

... Do you think the assumption that α_r and β_r do not vary across generations t is plausible? Can we test it?

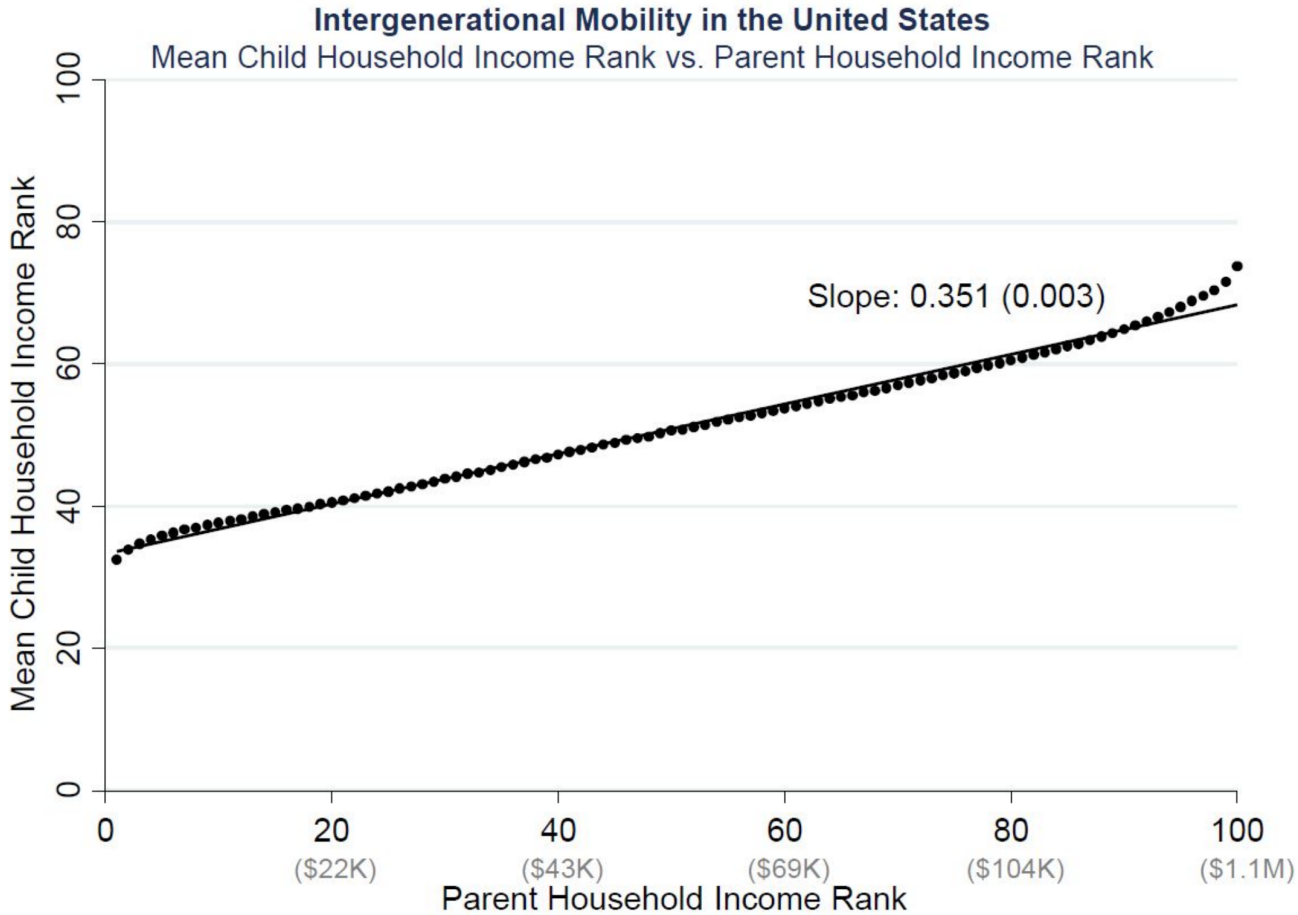


If blacks and whites had the same intergenerational mobility curve, inequalities should decrease fast from one generation to another.

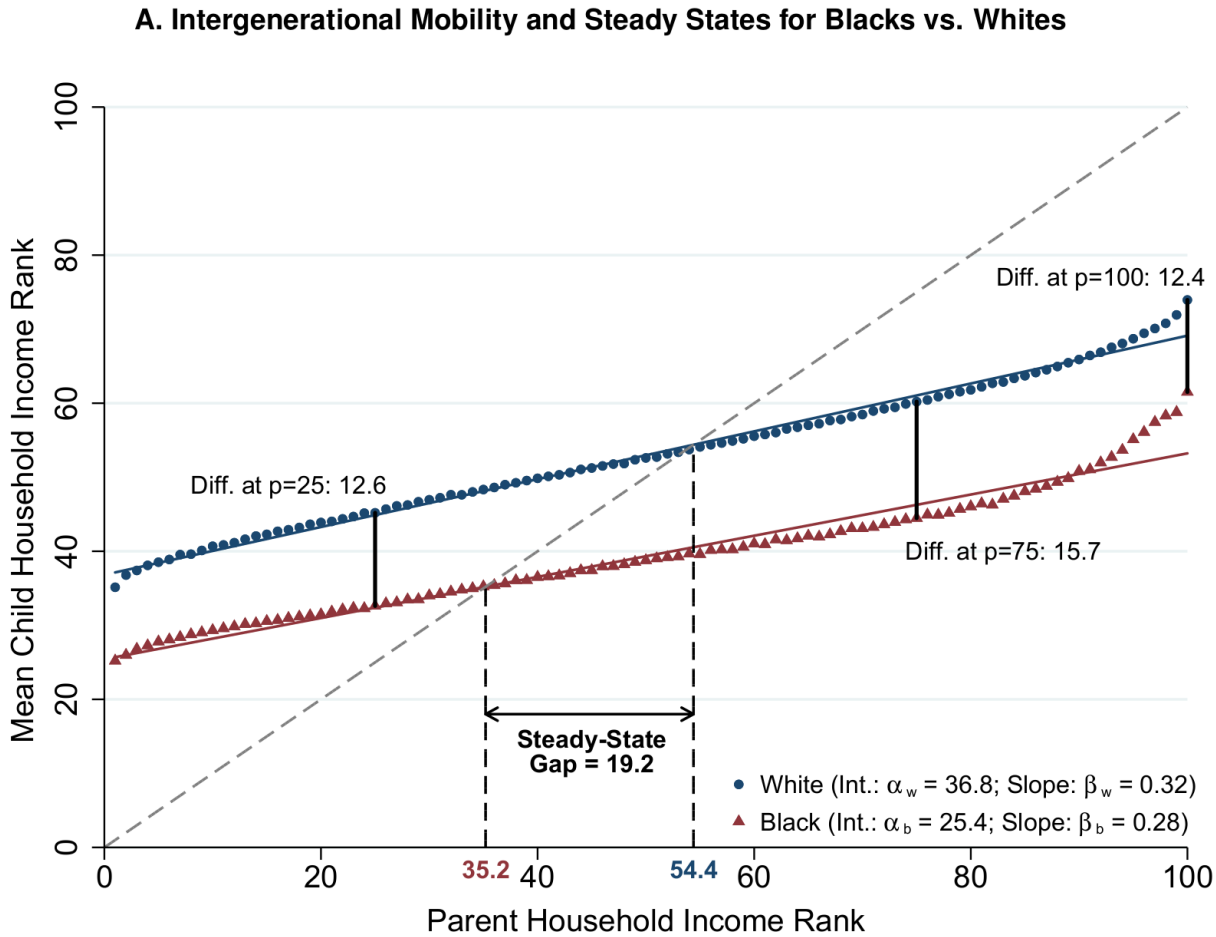


If they don't, then inequalities may persist indefinitely.

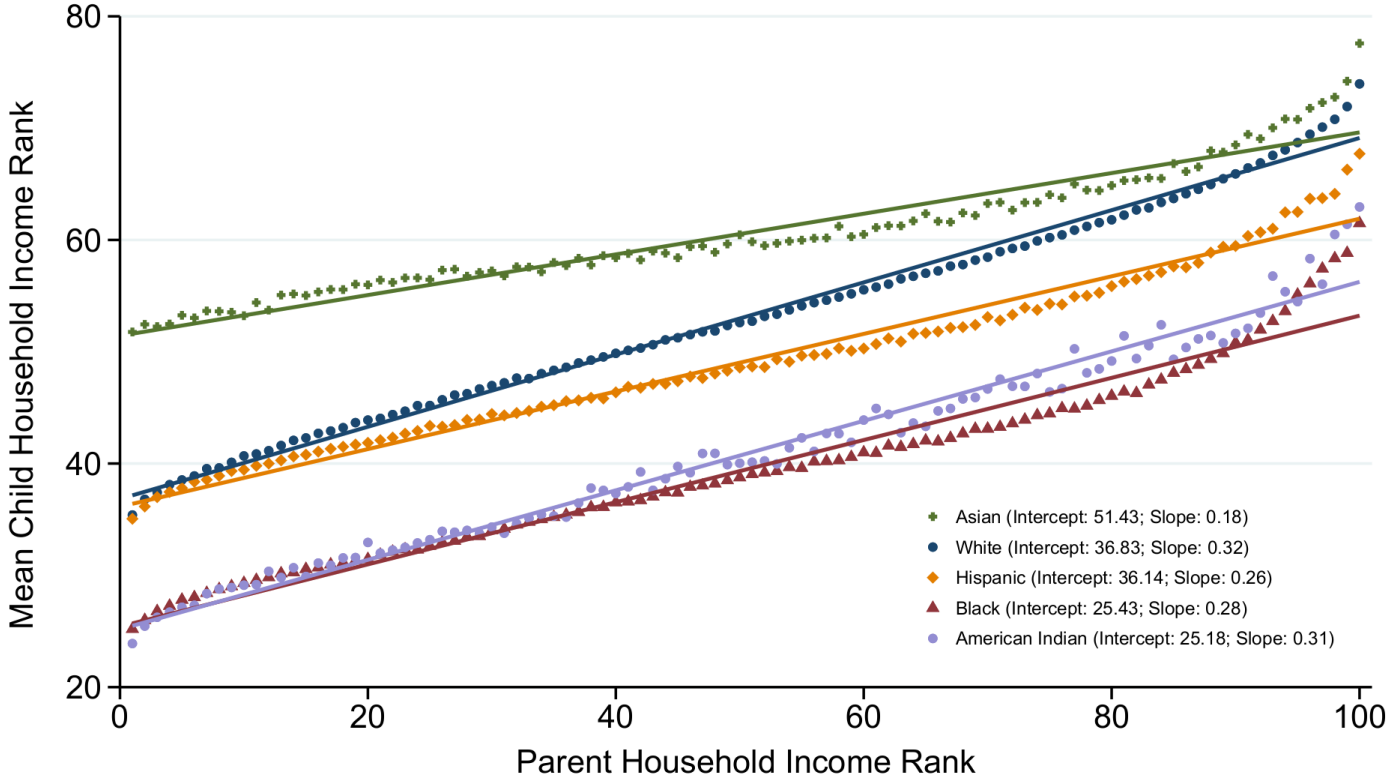
Let's now take a look at the real data...



Intergenerational mobility pooled across races is indeed approximately linear.

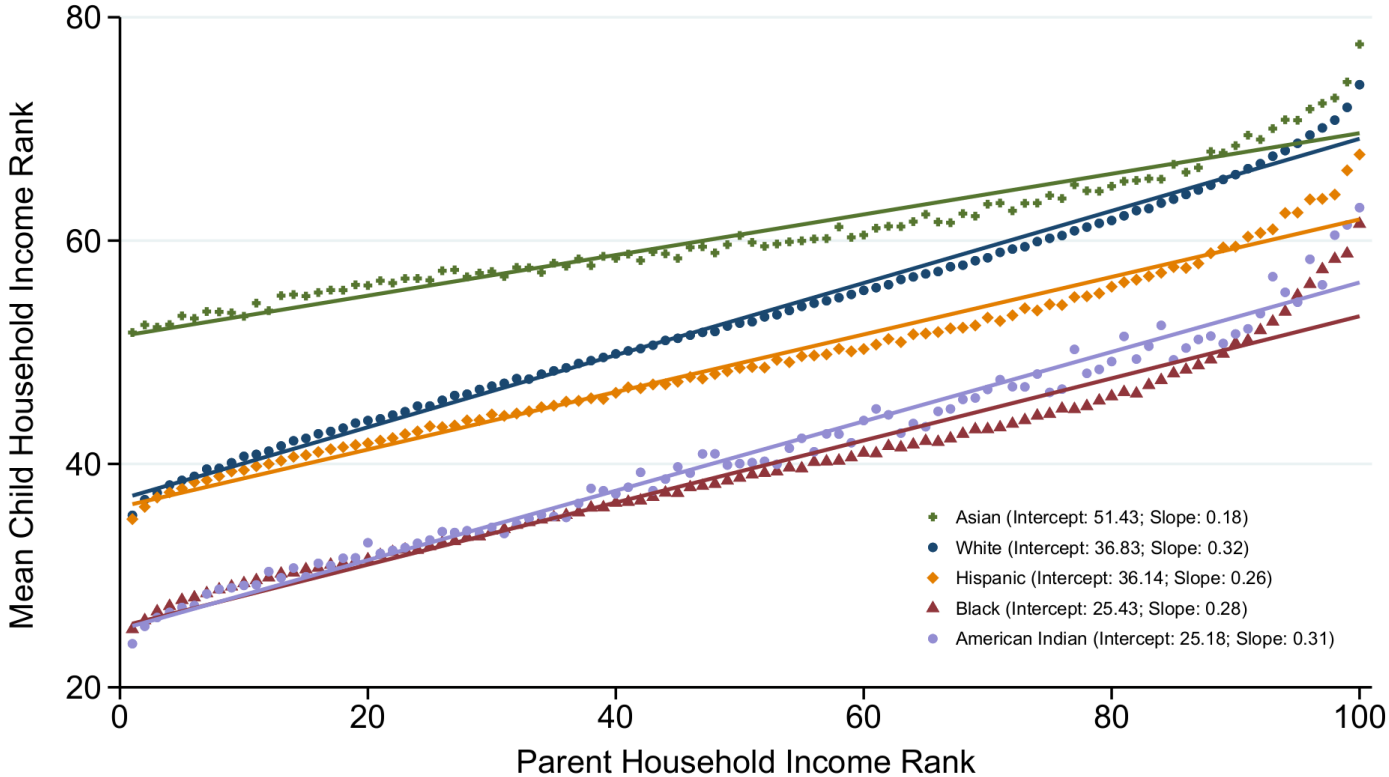


In practice, black and whites face indeed very different income mobility curves.



Intergenerational mobility relationship for Native Americans almost identical to the relationship for Black Americans.

Relationship for Hispanic Americans closer to what you see for whites than what you see for Black Americans.



Relationship for Asian-Americans much flatter: for these kids, parental income barely predictive of outcomes.

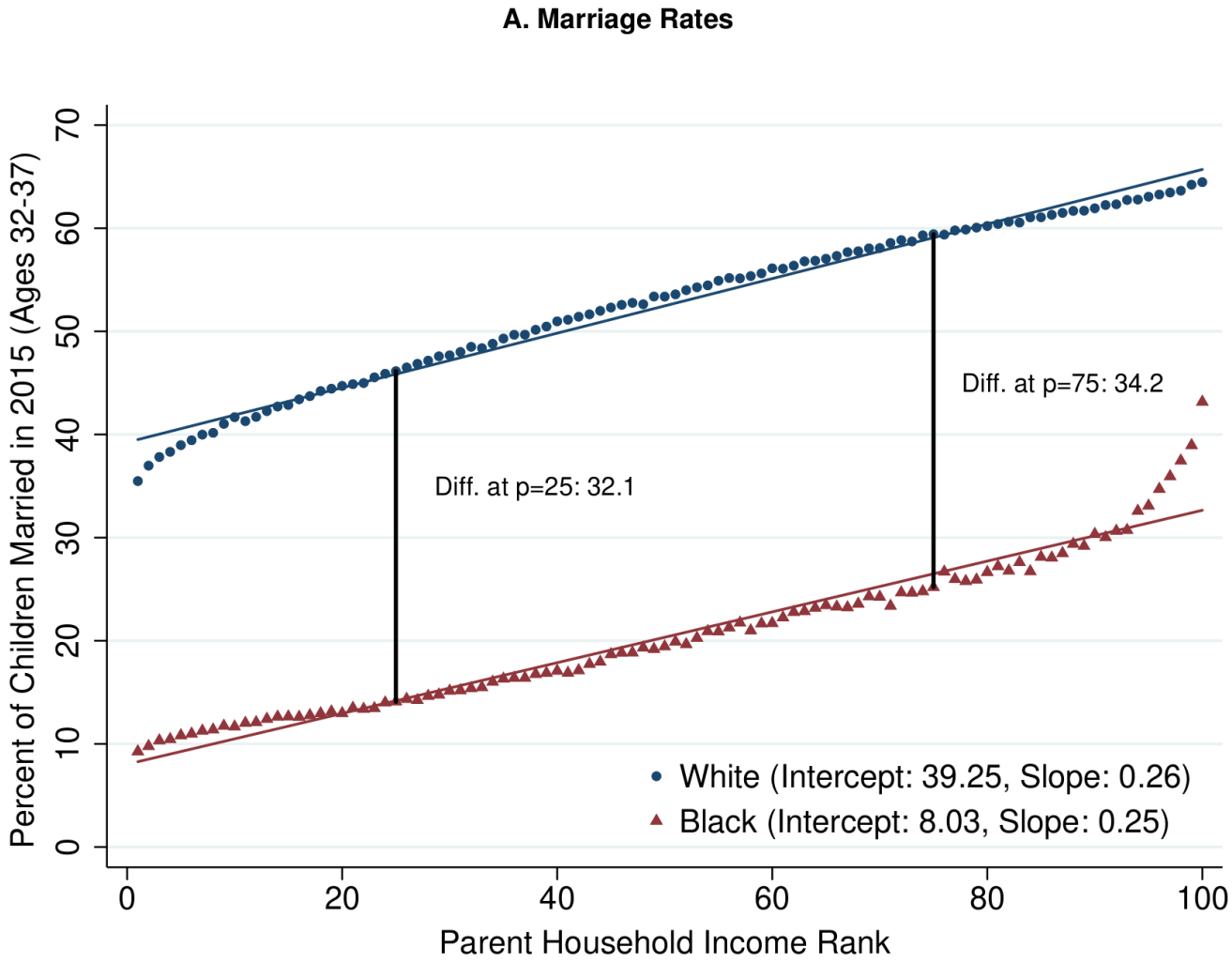
Relationship for Asian-Americans closer to that of whites if restricted to children with mothers born in the US.

Why do we observe intergenerational income mobility gaps?

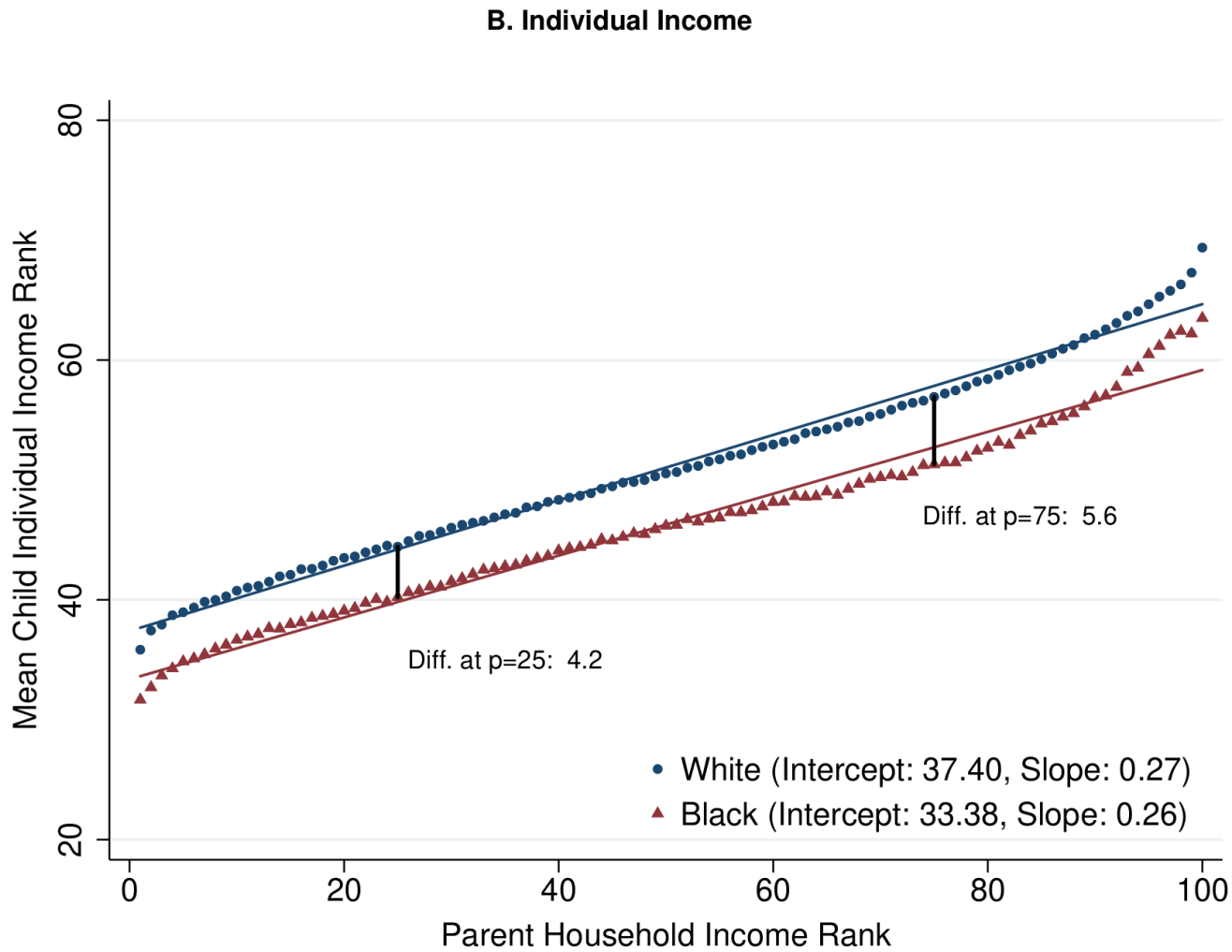
Chetty et al. (2020) consider several mechanisms:

- Marriage rates: Blacks marry less than whites, so they have lower household income levels (mechanical explanation).
- Employment rates: Blacks are more likely to be out of the workforce.
- Incarceration rates: Blacks are more likely to be incarcerated.
- Other environmental factors: family characteristics (marriage status, wealth, education, and neighborhood of parents)

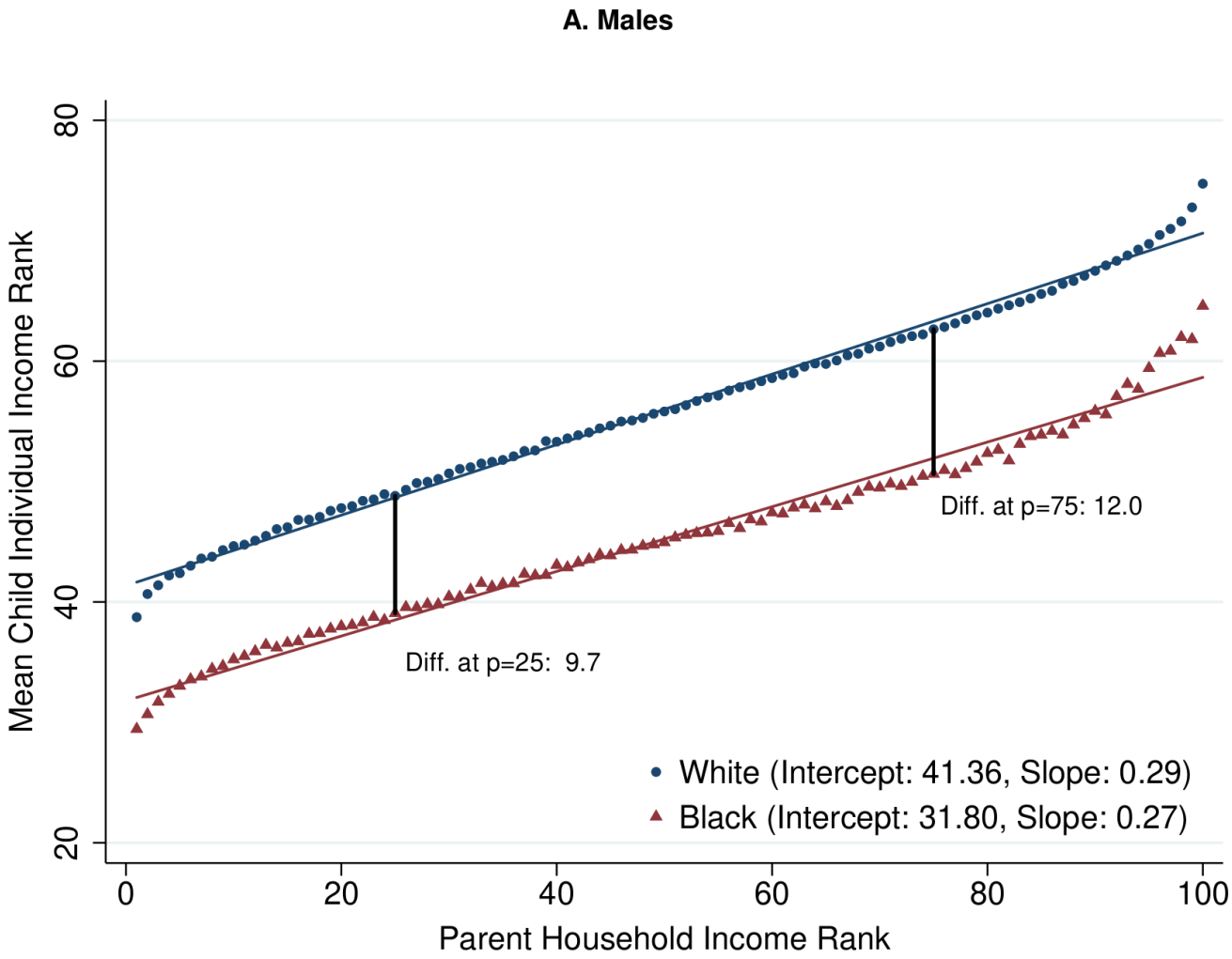
Overall, one striking pattern is the major differences between black men-women regarding intergenerational mobility.



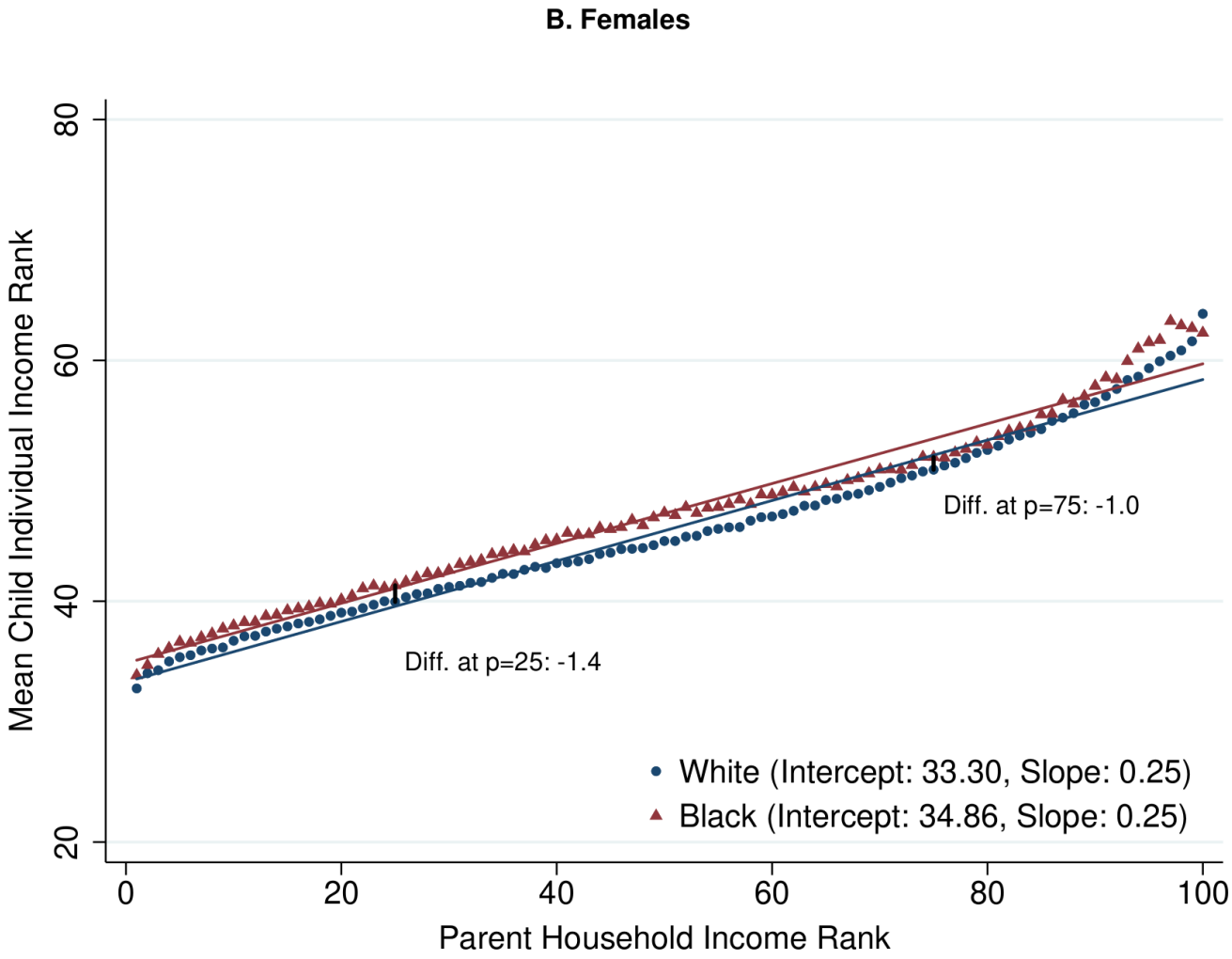
Whites are much more likely to be married.



But the racial gap persists even if we focus on individual (rather than household) income.

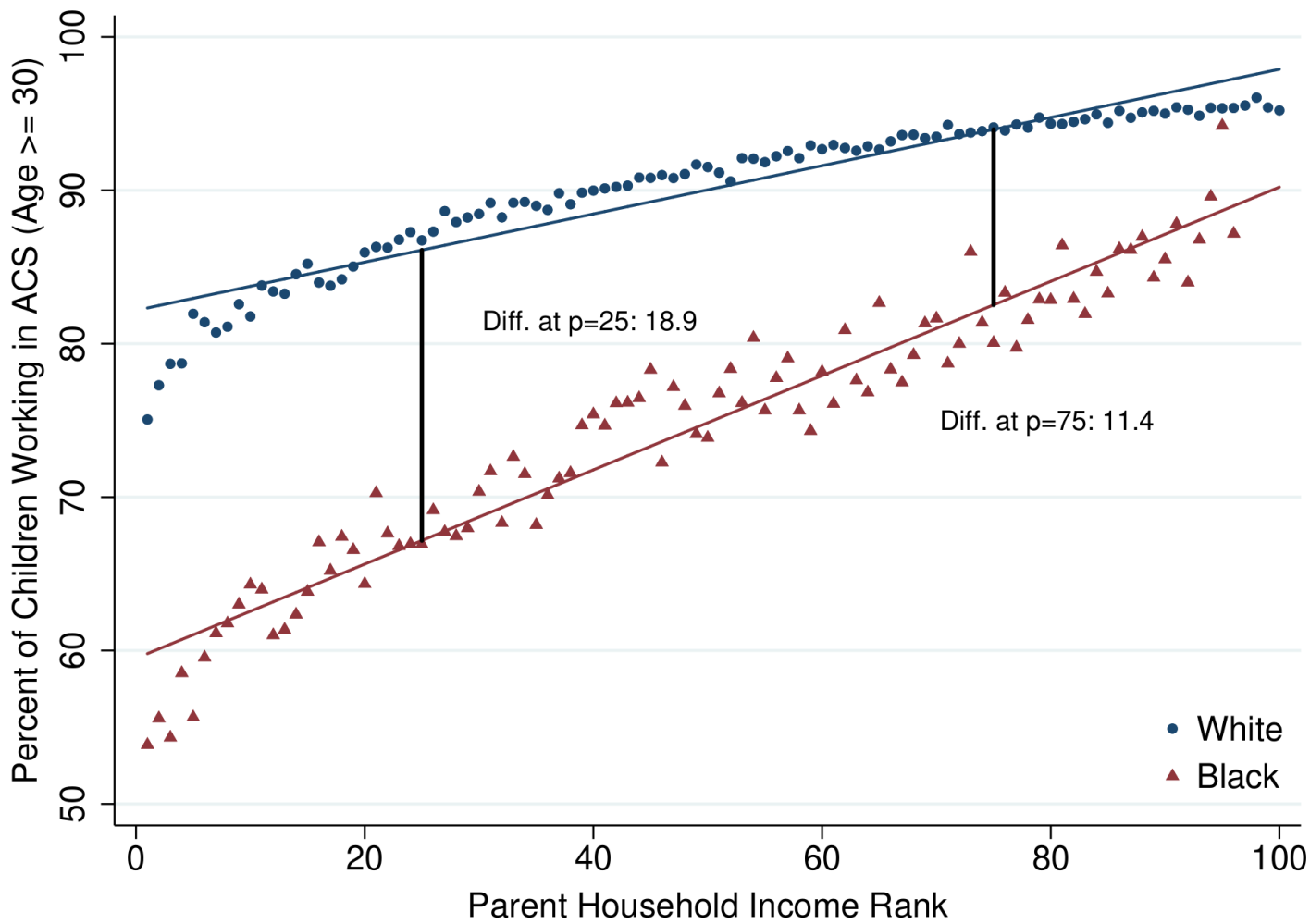


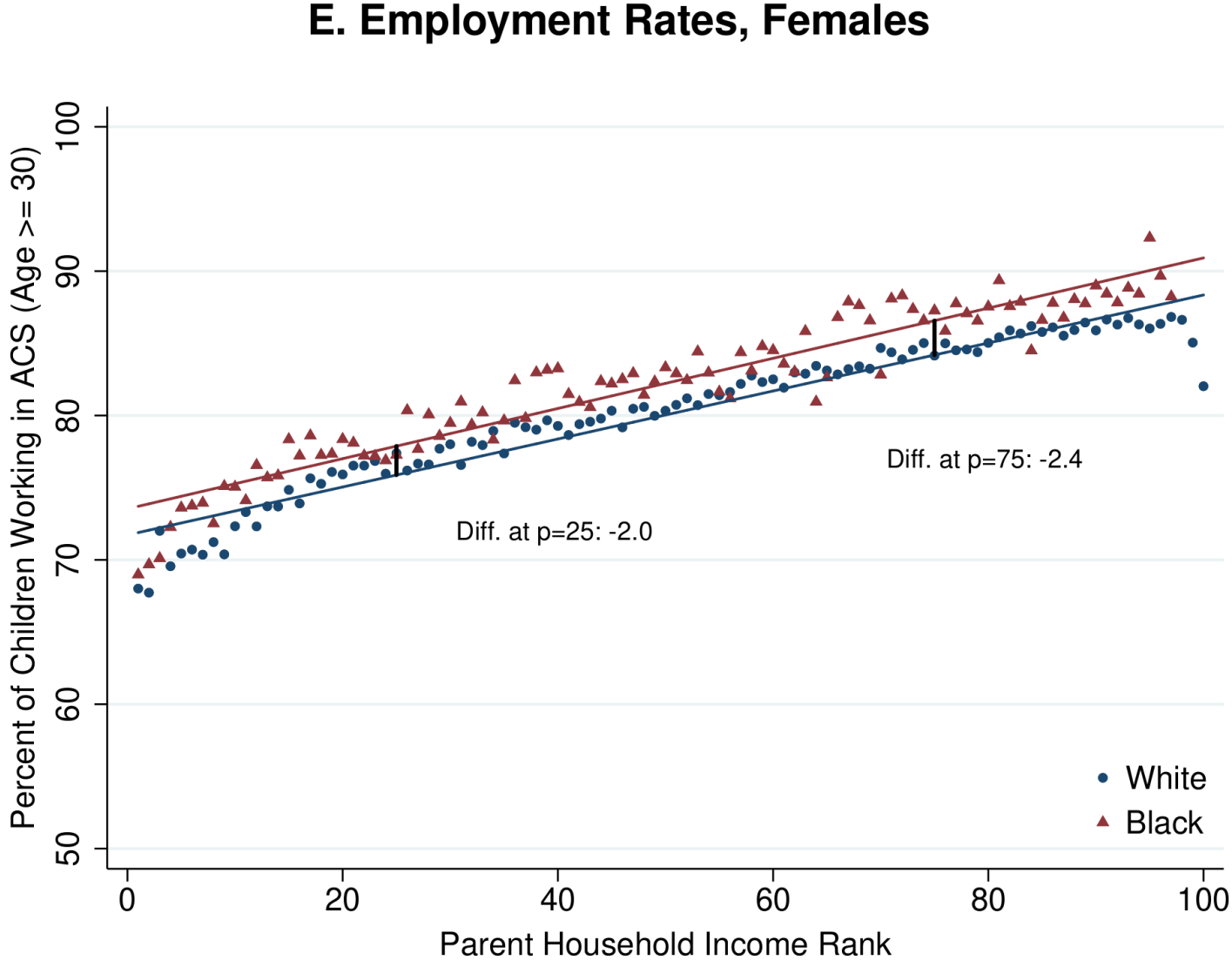
The gap is entirely driven by males.

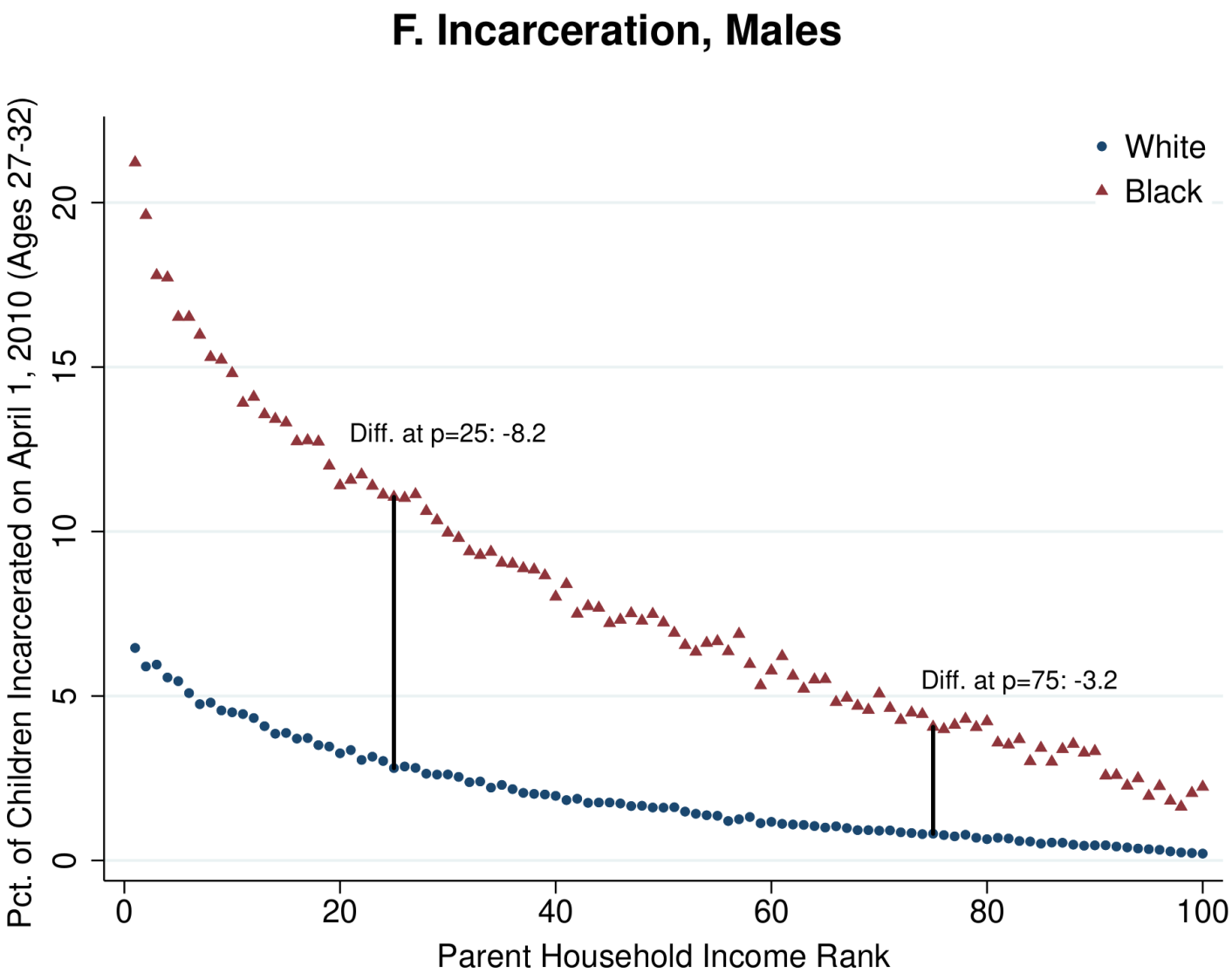


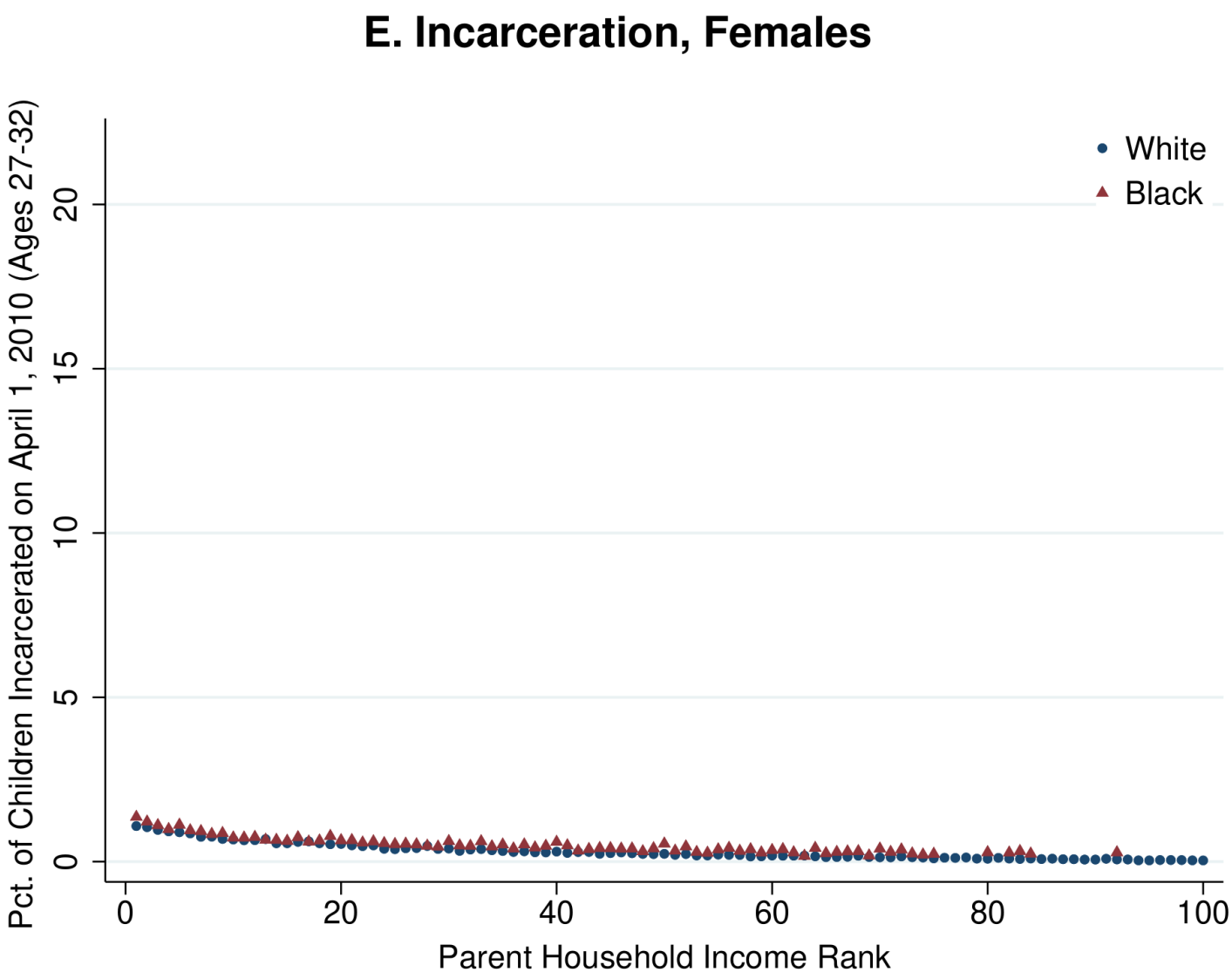
Black women actually have slightly higher intergenerational mobility prospects than white females.

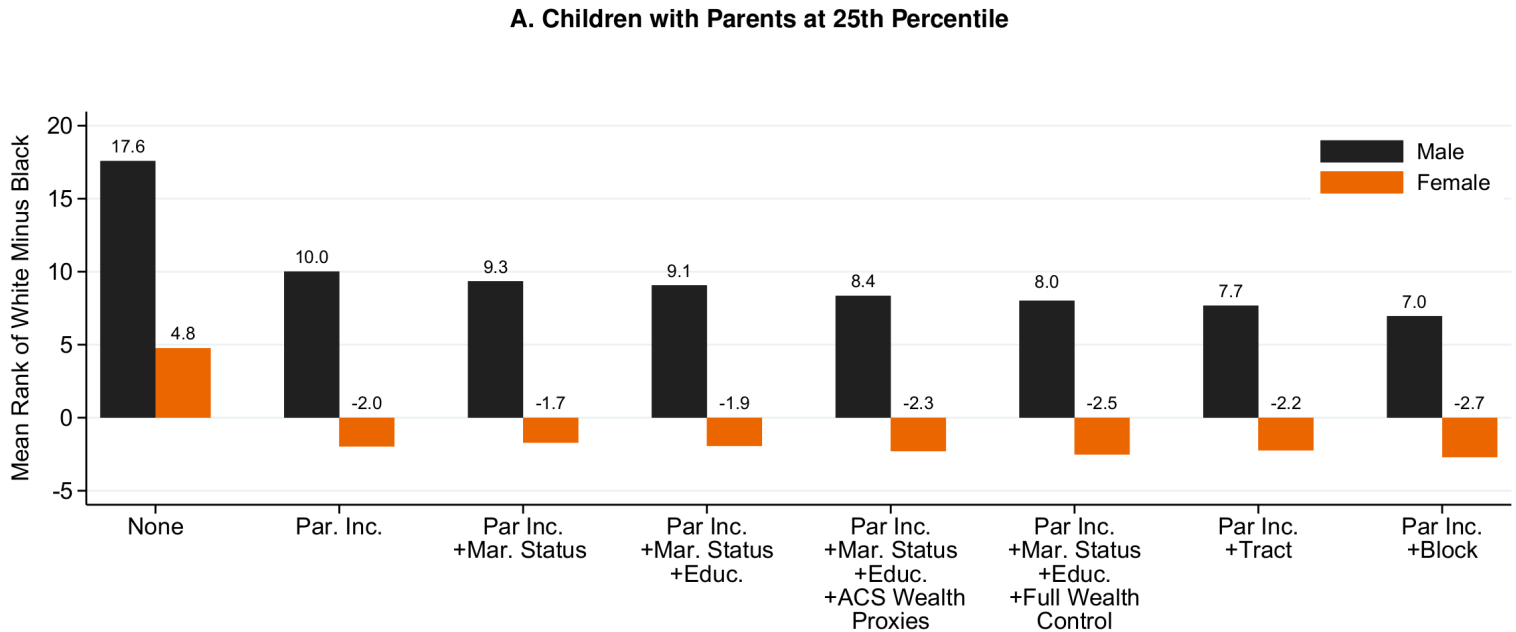
We observe similar patterns for employment and incarceration rates...


F. Employment Rates, Males
Percent of Children Working in ACS (Age >= 30)
50
60
70
80
90
100
0
20
40
60
80
100
Parent Household Income Rank
Diff. at p=25: 18.9
Diff. at p=75: 11.4
White
Black

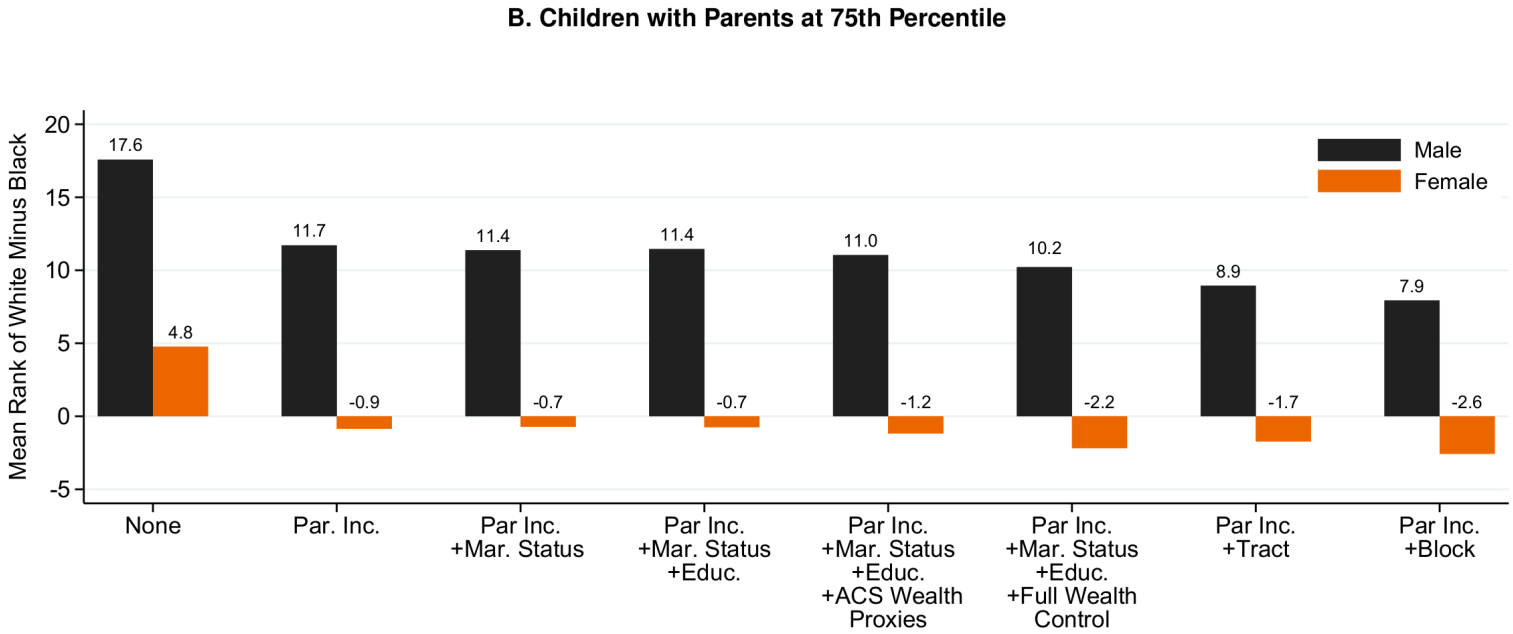

E. Employment Rates, Females
Percent of Children Working in ACS (Age >= 30)
50
60
70
80
90
100
0
20
40
60
80
100
Parent Household Income Rank
Diff. at p=25: -2.0
Diff. at p=75: -2.4
White
Black


F. Incarceration, Males
White
Black
Diff. at p=25: -8.2
Diff. at p=75: -3.2
Pct. of Children Incarcerated on April 1, 2010 (Ages 27-32)
0
5
10
15
20
0
20
40
60
80
100
Parent Household Income Rank


E. Incarceration, Females
White
Black
Pct. of Children Incarcerated on April 1, 2010 (Ages 27-32)
0
5
10
15
20
0
20
40
60
80
100
Parent Household Income Rank



Other family characteristics and geography explain part of the intergenerational mobility gap but do not bridge it.



Other family characteristics and geography explain part of the intergenerational mobility gap but do not bridge it.

What's Next?

We have documented large socio-economic, racial, and gender gaps in the US and worldwide.

Inequality between groups already provides reasonable grounds to study diversity.

But what if we were to reduce these gaps?

How would this affect economic and political outcomes we might care about?

Chetty, R., Hendren, N., Jones, M. R., and Porter, S. R. (2020). Race and economic opportunity in the united states: An intergenerational perspective. *The Quarterly Journal of Economics*, 135(2):711–783.

Tips for the Exam

Understand why inequality is, in part, an intergenerational problem.

Understand the measurement approach to intergenerational mobility.

In particular, be able to read and understand figures with intergenerational rank specifications.

Understand the factors driving intergenerational mobility gaps.